Name: _____

Instructions: All solutions should be prepared carefully and neatly. Any questions that ask you to explain or justify should have answers written in complete sentences. All solution sets shall be completed on this packet and handed in at the beginning of class on the due date below. Careless presentation (e.g. bad handwriting, pen scribbles, doodles, wasted space, etc) will result in a deduction of points at my discretion. If the question asks to submit a graph or plot, be sure to print the plot and include it in your final lab report. Use a PENCIL and if you make a mistake, use an eraser. Each student should prepare their own solutions that demonstrate their own comprehension of the material. Submitted work that does not demonstrate clearly the process by which one arrived at the answer will not receive credit of any kind. Academic dishonesty will not be tolerated.

Desmos Lab I

MATH 182 – Calculus II

Due: Tuesday, Dec 9th

Read: Section 10.3 and 10.8

Problem Number	Available Points	Your Points
1	10	
2	10	
Total	20	

1. Convergence of p-Series: We explore the convergence and divergence of p-series using Desmos. We will analyze the behavior of individual terms and then examine the behavior of partial sums.

   (a) **Understanding p-series.**

      i. Open a new graph in Desmos and save it as "p-Series Convergence."

      ii. The p-series is defined by $\sum_{n=1}^{\infty} \frac{1}{n^p}$. Enter the function $f(n) = \frac{1}{n^p}$ into Desmos.

      iii. Create a slider for p varying from 0.5 to 3. What happens to the terms of the series as n increases when $p = 0.5$?

      iv. What happens to the terms of the series as n increases when $p = 2$?

      v. Determine the approximate value of p at which the series transitions from divergence to convergence.

   (b) **Understanding partial sums.**

      i. Define the partial sums S_N in Desmos as:

$$S_N = \sum_{n=1}^{N} \frac{1}{n^p}, \quad N = 1, 2, \ldots, 50.$$

      Using Desmos, plot the partial sums S_N as a function of N.

      ii. How does S_N behave for $p = 0.5$ and for $p = 2$?

      iii. Explain how this behavior corresponds to convergence or divergence of the p-series.

      iv. When you submit this assignment on Desmos, share the url to this graph.

2. Taylor Series Exploration: We visualize Taylor series approximations and analyze their accuracy. Specifically, we look at how the degree n and the distance from the center affect the error.

   (a) **Visualizing a Taylor Polynomial.**

      i. Open a new graph in Desmos and save it as "Taylor Series Exploration."

      ii. Choose one of the following functions: $f(x) = e^x$, $f(x) = \sin(x)$, or $f(x) = \ln(1+x)$. Plot your chosen function in Desmos.

      iii. Add its Taylor polynomial of degree n centered at $x = 0$:

$$T_n(x) = \sum_{k=0}^{n} \frac{x^k}{k!}$$

      (Note: If you chose $\sin(x)$ or $\ln(1+x)$, adjust the formula to match the correct Taylor series).

      iv. Create a slider for n ranging from 1 to 100. How does increasing n affect the accuracy of the approximation?

      v. For which values of x does the Taylor polynomial approximate the function well?

      vi. Compare $T_3(x)$, $T_5(x)$, and $T_{10}(x)$ on the same graph.

(b) **Error Analysis.**

i. Define the approximation error:

$$E_n(x) = |f(x) - T_n(x)|$$

Plot $E_n(x)$ in Desmos for different values of n.

ii. How does the error change as n increases?

iii. How far from $x = 0$ does the Taylor polynomial remain a good approximation?

iv. When you submit this assignment on Desmos, share the url to this graph.